G-Matrix Systems
Integrated System Design

RAMAC® Solutions:

HOP Ecosystem

A Smarter Way To Track - *Anything*

RAMAC™ Solutions: HOP Ecosystem
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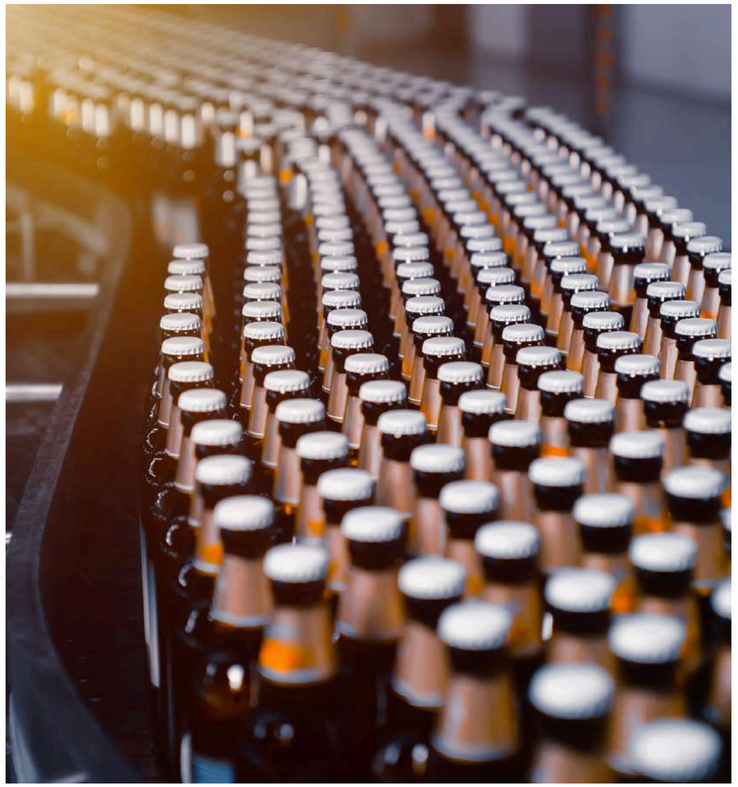
RAMAC® Solutions: HOP Ecosystem



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RAMAC® Solutions: HOP Ecosystem

Smarter Tracking

for Smarter Operations

The RAMAC™ HOP Ecosystem from G-Matrix Systems is a powerful, next-generation asset tracking platform that leverages Bluetooth™ Low Energy (BLE) technology to deliver accurate, real-time visibility of goods, equipment, and personnel, indoors, outdoors and in transit.

Traditional GPS or Wi-Fi-based tracking systems can be expensive, infrastructure-heavy, and limited in enclosed environments like warehouses, trucks, or factories.

G-Matrix Systems overcomes these limitations by creating a self-forming Bluetooth™ mesh network using a combination of RAMAC™ HOP Relays, GSM Cards, RAMAC™ HOP Repeaters and RAMAC™ HOP Tags.

This unique architecture allows the RAMAC™ HOP ecosystem to reach the cloud via the nearest connected devices and transmit asset data in real time.

The RAMAC™ HOP Ecosystem gives your business the ability to "see" the assets in real-time, even when they are inside metal trucks, in transit, or stored in remote yards. This is especially valuable in logistics, construction, healthcare, manufacturing, and other industries where things move fast, and often get lost.

The RAMAC™ HOP Ecosystem is completely scalable. You can start small tracking a few key assets, and expand gradually as your needs grow.

No complex cabling, no wired gateways, no extensive IT resources. It’s essentially plug-and-play solution with rapid deployment and low total cost of ownership.

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RAMAC™ HOP Ecosystem

RAMAC® HOP Relay

he RAMAC™ HOP Relay links the HOP ecosystem to the cloud, transmitting data from RAMAC™ Repeaters and Tags. Its rugged, energy-efficient design delivers dependable performance in any environment.

RAMAC® HOP Repeaters

High and Low Volume

The RAMAC™ HOP Repeaters extend your network by relaying signals between Relays and Tags, enabling reliable real-time tracking indoors or outdoors. Available in compact low-volume and high-volume variants, they offer durable, low-power performance for scalable asset monitoring across small or large areas.

RAMAC® GSM Card

The RAMAC™ GSM Card is a smart, wearable BLE badge that tracks staff and assets while combining safety, communication, and access control. Built for reliability, it performs in demanding environments like construction sites, healthcare facilities, and offices, or on the move.

RAMAC® HOP Tags

The RAMAC™ HOP Tag is a compact, rugged device for real-time asset and environmental monitoring. Seamlessly integrated within the RAMAC™ HOP ecosystem, it delivers reliable indoor and outdoor performance with long battery life.

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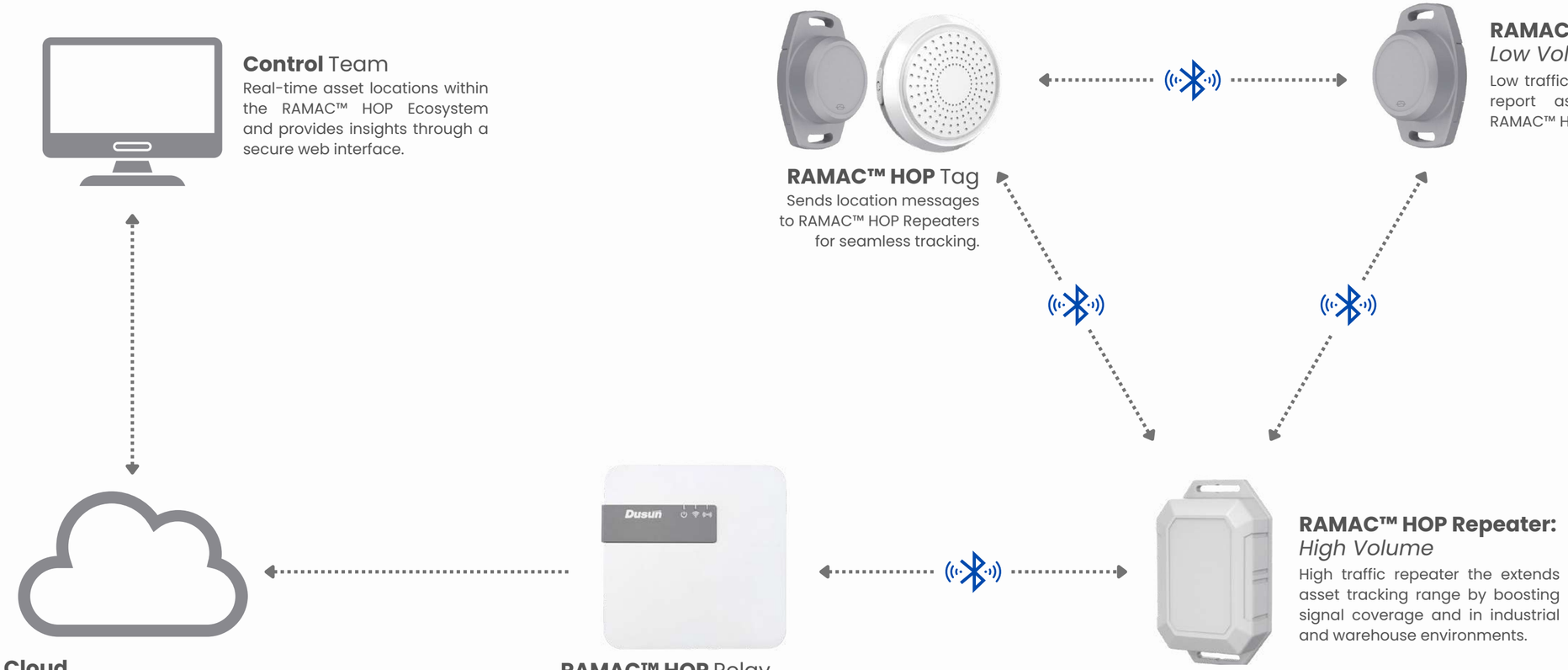
RAMAC® HOP Ecosystem Overview - Controlled Environment



Control Team
Real-time asset locations within the RAMAC™ HOP Ecosystem and provides insights through a secure web interface.

RAMAC™ HOP Tag
Sends location messages to RAMAC™ HOP Repeaters for seamless tracking.

RAMAC™ HOP Repeater:
Low Volume
Low traffic repeater monitors and report asset location in the RAMAC™ HOP ecosystem.

Cloud
Processes and stores device data, delivering tracking and RAMAC™ HOP Ecosystem insights.

RAMAC™ HOP Relay
Receives traffic from RAMAC™ HOP Tags and Repeaters and connects to the cloud for real-time data flow.

RAMAC™ HOP Repeater:
High Volume
High traffic repeater the extends asset tracking range by boosting signal coverage and in industrial and warehouse environments.

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RAMAC® HOP Ecosystem Overview - In Transit

Control Team

Real-time asset locations within the RAMAC™ HOP Ecosystem and provides insights through a secure web interface.

RAMAC™ HOP Tag

Sends location messages to RAMAC™ HOP Repeaters for seamless tracking.

RAMAC™ HOP Repeater:
Low Volume

Low traffic repeater monitors and report asset location in the RAMAC™ HOP ecosystem.

Cloud

Processes and stores device data, delivering tracking and RAMAC™ HOP Ecosystem insights.

RAMAC™ GSM Card

Connects RAMAC™ HOP Dots, Trackers and Repeaters to the cloud for real-time data flow.

RAMAC™ HOP Repeater:
High Volume

High traffic repeater the extends asset tracking range by boosting signal coverage and in industrial and warehouse environments.

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RAMAC® **Solutions:** HOP Ecosystem

RAMAC™ HOP Relay

Where Smart Infrastructure Begins.

Connect. Track. Scale.

The RAMAC™ Bluetooth™ HOP Relay is a compact, cost-effective gateway that connects your RAMAC™ Bluetooth™ Trackers and RAMAC™ Repeaters to the cloud enabling real-time data flow across the RAMAC™ Bluetooth™ HOP ecosystem.

Built for scale and seamless integration, it's ideal for asset tracking, sensor data collection, and environmental monitoring in industrial, commercial, and smart building environments.

Key Features & Benefits

- **HOPtimised:** Seamlessly connects with RAMAC™ Repeaters.
- **Dual Connectivity:** Wi-Fi and Ethernet for flexible deployment.
- **Low-Power BLE 5.0:** Long-range, energy-efficient Bluetooth comms.
- **Edge Processing:** Reduces latency and cloud load.
- **Secure & Scalable**: Encrypted and ready for large networks.
- **Compact Design:** Ideal for warehouses, factories, hospitals, and smart spaces.
- **Plug & Play:** Fast, reliable BLE-to-cloud integration.

Ideal Industry Use:

Industrial | Commercial/ Business | Smart Buildings | Manufacturing/ Factory

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RAMAC® **Solutions:** HOP Ecosystem

RAMAC™ HOP Relay

Specifications

- **Dimensions:** 95x95x30mm
- **CPU:** ESP32
- **System:** FreeRTOS
- **RAM:** 520KB
- **Flash:** 4MB
- **Reset:** Factory reset button

Connectivity

- **SIM:** 1x Micro SIM slot
- **Ethernet:** 1x 10/100Mbps WAN/LAN variable
- Supports LTE Cat M1
- **Antenna:** LTE FPC Antenna, Zigbee or PCB Antenna
- **Wireless Protocol:** Wi-Fi 2.4G, Zigbee 3.0 OR Bluetooth 5.2

Operating Conditions

- **Installation:** Flat surface or ceiling
- **Operating Temp.:** -10°C to 55°C
- **Storage Temp.:** -20°C to 65°C
- **Operating Humidity:** 10% to 90%

Power Supply

- **USB:** 5V/2A type-C power supply
- **Green LED:** when powered ON
- **Red LED:** BLE/Zigbee LED flash for signal
- **Blue LED:** Wi-Fi LED on when connected to network

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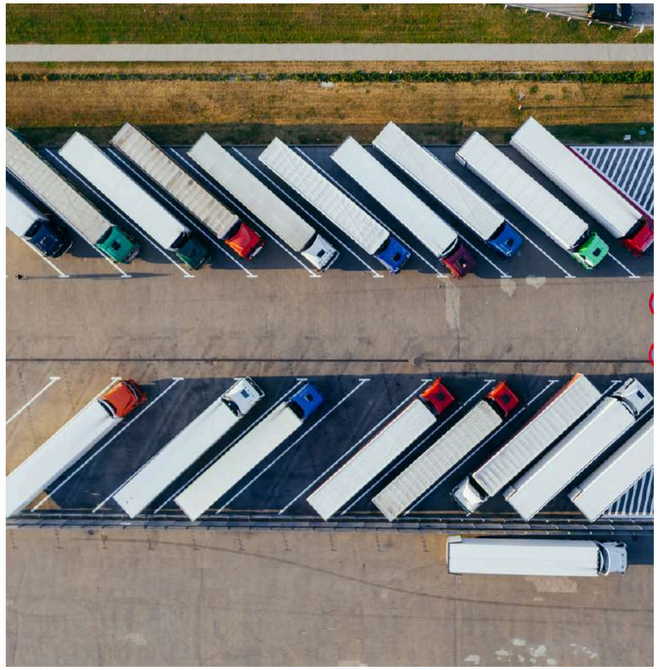
RAMAC® **Solutions:** HOP Ecosystem

RAMAC™ GSM Card



Safety You Can Wear.

Real-time awareness, anytime, anywhere.

The RAMAC™ GSM Card is a smart, wearable badge that combines ID card functionality with Bluetooth Low Energy (BLE) technology.

Designed for staff monitoring and location awareness of people and assets, it provides an all-in-one solution for safety, communication, and access control.

Built with reliability in mind, the RAMAC™ GSM Card is suited to demanding environments such as construction sites, healthcare facilities, and offices.

Key Features & Benefits

- **HOPtimised:** Seamlessly connects with RAMAC™ HOP Repeaters and Relays.
- **CR80 ID Card Support** with side insertion.
- **Low-Power BLE 5.0:** Long-range, energy-efficient Bluetooth comms.
- **RFID/NFC** dual sided sensing.
- **Dual Buzzer and LED Alerts** for instant response.
- **Compact Design:** Ideal for warehouses, factories, hospitals, offices, schools and smart spaces.
- **Flexible Wear** with horizontal or vertical lanyard options

Ideal Industry Use:

Fleet Driver | Employee/ Crowd/School | Smart Buildings/ Construction | Hospitals/ Patients

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RAMAC® **Solutions:** HOP Ecosystem

RAMAC™ GSM Card

Specifications

- **Dimensions:** 98x65x8mm
- **Weight:** 35g
- **Colour:** White Body, Red Button
- **RFID:** HF | LF/UHF
- **Suitable Card Size:** CR80 IC Card
- **IP Rating:** IP66
- **Material:** PC

Connectivity

- **Connectivity:** BLE 5.0
- **Sensor:** Accelerometer / Buzzer

Operating Conditions

- **Installation:** Lanyard
- **Operating Temp.:** -20°C to 60°C
- **Transmission Range:** 150m unobstructed

Power Supply

- **Battery Capacity:** 800mAh, non-replaceable
- **Battery Lifespan:** 4 years*
- **LED:** Varied LED notification

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RAMAC® **Solutions:** HOP Ecosystem

RAMAC™ HOP Repeaters - High & Low Volume

Extend. Track. Protect.

Built to Extend Your Tracking Range.

Extend your RAMAC™ HOP Ecosystem with the RAMAC™ Repeaters High & Low Volume for reliable asset and staff tracking.

The RAMAC™ Repeater High Volume delivers long-range coverage, up to 150 m unobstructed, long battery life and industrial-grade durability with flexible mounting and optional sensors for temperature, humidity, and water detection.

The RAMAC™ Repeater Low Volume is a compact, portable repeater for indoor or outdoor asset monitoring. Easy to install without tools, it also provides an unobstructed 150 m range for scalable, real-time tracking.

Key Features & Benefits:

- **Long-Range BLE Coverage:** Up to 150 m for reliable tracking in large spaces.
- **Extended Battery Life**: Provides long-term low-maintenance operation.
- **Durable & Versatile:** Rugged design for harsh environments.
- **Flexible Installation:** Adaptable installation for wide range of surface.
- **Smart Monitoring:** Optional additional sensors that enable temperature, humidity, water detection

Ideal Industry Use:

Warehouse | Manufacturing/ Factory | Storage | Construction

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RAMAC® **Solutions:** HOP Ecosystem

RAMAC™ HOP Repeater - High Volume

Specifications

- **Dimensions:** 123x73x34mm
- **Colour:** Light Grey
- **Shock Resistance:** IK07
- **IP Rating:** IP67
- **Material:** PC + ASA
- **Messages:** 5000 per day

Operating Conditions

- **Installation:** Screw/ZipTie/3M Adhesive Tape
- **Operating Temp.:** -40°C to 85°C

Connectivity

- **Connectivity:** BLE 5.0
- **Chipset:** Nordic nRF52 series
- **Sensor:** 3-Axis Sensor / Hall Sensor / Leak Detection Sensor / Temperature + Humidity Sensor
- **Transmission Range:** Up to 150m without obstacles

Power Supply

- **Battery Capacity:** 2*AA ER18505 8000mAh Replaceable Battery

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RAMAC™ HOP Repeater - Low Volume

Specifications

- **Dimensions:** 70x46x21mm
- **Weight:** 33.5g
- **Colour:** Gray
- **Shock Resistance:** IK08
- **IP Rating:** IP67
- **Material:** PC + ABS
- **Message:** 500 per day

Operating Conditions

- **Installation:** Hang/Sticker/Screw
- **Operating Temp.:** -20°C to 60°C
- **Humidity:** 0% - 95% non-condensing

Connectivity

- **Connectivity:** BLE 5.0
- **Button:** Inner Mechanical Button
- **LED:** Single Red LED
- **Sensor:** Accelerometer Sensor / Optional Magnetic Switch
- **Transmission Range:** Up to 150 meters without obstacles.

Power Supply

- **Battery Capacity: 1000m**Ah Replaceable Battery
- **Battery Lifespan:** 3+ Years with default settings

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RAMAC® **Solutions:** HOP Ecosystem

RAMAC™ HOP Tags

Built to Track, Ready for Anything.

Your assets, tracked with precision and reliability.

The RAMAC™ HOP Tags deliver reliable real-time asset tracking and monitoring with versatile form factors to suit different needs.

Using Bluetooth 5.0, they provide connectivity of up to 150m and feature waterproof designs (IP65/IP67) built for demanding environments. Easy installation options allow for quick battery replacement, while the optional integrated sensors (temperature, humidity, barometric, and a 3-axis accelerometer) enable both environmental and motion monitoring.

The RAMAC™ HOP Tags ensure dependable and adaptable location tracking across industries.

Key Features & Benefits

- 150m unobstructed asset tracking.
- Rugged design.
- Motion detection via built-in accelerometer.
- Temperature, humidity, and barometric monitoring.
- Flexible mounting and quick installation.
- Seamless integration into the RAMAC™ Bluetooth HOP ecosystem for scalable networks.

Ideal Industry Use:

Warehouse & Inventory

Vehicle and Fleet Monitoring

Personnel Tracking

Healthcare Equipment Monitoring

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RAMAC® **Solutions:** HOP Ecosystem

RAMAC™ HOP Tags

Specifications

- **Dimensions:** 70x46x21mm / 48x48x16mm
- **Weight:** 33.5g (with Battery) / 26g (with Battery)
- **Colour:** Gray / White
- **IP Rating:** IP67 / IP65
- **Material:** ABS + PC / ABS + PC & TPU
- **Messages:** 160 messages per day
- **Form Factor:** application dependent

Operating Conditions

- **Installation:** Hang/Stick/Screw/Neck Chain
- **Operating Temp.:** -20°C to 60°C
- **Storage Temp.:** -20°C to 60°C (without battery)
  10°C to 25°C (with battery)
- **Humidity:** 0% to 95% (non-condensing)

Power Supply

- **Battery:** CR2477 - 1000mAh (replaceable)

Connectivity

- **Bluetooth:** BLE 5.0
- **Sensor:** Optional: Temperature & Humidity sensor / Hall effect sensor / Accelerometer sensor / Barometric pressure sensor
- **Range:** Up to 150m unobstructed

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RAMAC® HOP Ecosystem Industries

Hospitality and Entertainment

- Indoor Navigation
- Point of Interest
- Flow Analysis
- Usage Level Detection
- Smart Parking
- Shuttle Bus Tracking
- Hotel Employee Safety

Hospitals and Healthcare

- Staff Protection
- Patient Visibility and Care
- Medical Equipment Management
- Environmental Monitoring

Education and Higher Learning

- Campus Safety
- School Bus Tracking
- Classroom Occupancy
- Environmental Monitoring

Facilities Management

- Asset Tracking
- Energy Management
- Room Occupancy
- Environmental Monitoring
- Usage Level Detection
- Smart Parking

Airports and Tourism

- Luggage Cart Detection
- Wheelchair Monitoring
- Equipment Management

Warehousing and Storage Management

- Asset Tracking
- Inventory Management
- Forklift Management
- Staff Protection
- Location Occupancy

Freight and Fleet Management

- Vehicle/Fleet Tracking
- Container & Goods Tracking
- Driver Safety & Identification
- Cold Chain Monitoring

Construction and Maintenance

- Worker Safety & Monitoring
- Tool & Equipment Tracking
- Asset Tracking
- Site Occupancy
- Personnel Tracking

Manufacturing and Factories

- Worker Safety & Monitoring
- Tool & Equipment Tracking
- Asset Tracking
- Location Occupancy
- Personnel Tracking
- Inventory Monitoring

Ocean Freight, Cargo and Harbours

- Precise tracking of container locations for billing and retrieval
- Repeaters embedded in roads
- Worker Safety & Monitoring
- Tool & Equipment Tracking

Automotive Supply Chain

- Tracker for precise location at factory and assembly
- Tool & Equipment Tracking
- Asset Tracking
- Location Occupancy
- Personnel Tracking

Logistics and Supply Chain

- Asset Tracking
- Inventory Management
- Staff Protection
- Location Occupancy

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Turnkey IoT Solutions

Hardware Development, Prototyping, Firmware Development, Database Design, Testing Integration and Web Integration.

Contact Us

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